Fiscal Year:	FY 2007	**Task Last Updated:**	FY 11/27/2007
PI Name:	Merfeld, Dan Ph.D.		
Project Title:	Influence of Sensory Integration on the Neural Processing of Gravito-Inertial Cues		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Boston	**State:**	MA
Zip Code:	02114	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-02
Start Date:	12/01/2003	**End Date:**	03/31/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date is 3/31/2007 per PI (11/06). NOTE: Funding reinstated to 11/30/2006, per J. Cohen (HQ, 8/2006)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Lionel, Zupan Ph.D. (MEEI)		
Grant/Contract No.:	NNJ04HB01G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Our sense of spatial orientation results from a complex set of neural processes of sensory integration; these processes utilize information from many different physiological systems. Evidence suggests that extended exposure to micro-gravity yields adaptive changes in these neural processes. For the most part, these adaptive changes appear functionally relevant for spaceflight but yield significant transient problems when astronauts return to gravitational environments (Earth, Mars, Moon, etc.). In-flight countermeasures that reinforce the normal sensory interactions encountered in a gravitational environment (e.g., rotations about any axis not aligned with gravity are accompanied by changes in the relative orientation of gravity) will almost certainly help astronauts function adequately when they transition to a gravitational environment. This will be important to pilot astronauts landing spacecraft as well as to crewmembers trying to rapidly exit a spacecraft. Furthermore, artificial gravity has been proposed as a countermeasure for numerous problems associated with spaceflight (including cardiovascular deconditioning, bone decalcification, etc.). One undesired side effect of artificial gravity will be additional alterations in these processes of sensory integration. For all of these reasons, it is essential that we understand the predominant sensory interactions that are altered by exposure to micro-gravity. Specifically, we proposed to investigate how rotational signals from the semicircular canals influence how humans process the ambiguous gravito-inertial force measured by the otolith 0rgans. We are now performing a series of ground experiments that seek a better understanding of sensory interactions between the semicircular canals and graviceptors. We propose to perform these investigations using several different measurement techniques including measures of reflexive eye responses as well as psychophysical measures of human perception.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In our studies, we are investigating how normal subjects process ambiguous sensory cues from the otolith organs, which measure gravity and linear acceleration. These basic science studies will lead to improved diagnostic testing for human patients suffering from balance and vestibular problems.
Task Progress:	As described in detail in the progress report, we successfully achieved the scientific objectives. Scientific goals/objectives: This project was originally a flight experiment that was scaled back to be a ground study shortly after the space shuttle accident. After this change in plans, our project goals were altered. Our new goal was to use psychophysics and eye movement measurements to investigate how humans utilize semicircular canal cues to help estimate the relative orientation of gravity here on Earth. We proposed a ground study to measure human responses (perceptual and eye movement) during combined canal-otolith stimulation on Earth. Our goal was to develop new tests that will allow us to investigate how canalotolith interactions adapt to micro-gravity exposure. Subjects were tested using protocols in which the subjects a) tilted, b) translated, or c) tilted & translated simultaneously. We expected the human responses to be consistent with those previously measured in monkeys, with canal cues essential for accurate estimation of gravity and linear acceleration. While many details are provided below and in our publications, these goals were all successfully achieved. As a brief overview: 1. We developed and published (Merfeld, 2003) a new hypothesis designed to explain spaceflight adaptation, the disorientation experienced in microgravity, and the motion sickness that sometimes occurs. 2. We successfully developed a new testing paradigm that allowed us to investigate human tilt and translation responses. The new testing paradigm (described in detail in Merfeld et al., 2005b) is ready to be used as part of a preflight/postflight investigation to study how astronauts adapt to the micro-gravity environment. 3. We successfully completed an extensive set of human studies in which we found that the mechanism used to evoke human translational VOR responses is different than the mechanism used to evoke monkey translational VOR responses. 4. These same studies showed that human perception and human action (i.e., the VOR) employ qualitatively different neural mechanisms, which emphasizes the importance of studying perception to complement our existing extensive understanding of the VOR. Publications Referenced: Merfeld D (2003) Rotation Otolith Tilt-Translation Reinterpretation (ROTTR) Hypothesis: A new hypothesis to explain neurovestibular spaceflight adaptation. Journal of Vestibular Research 13:309-320. Merfeld D, Park S, Gianna-Poulin C, Black FO, Wood S (2005a) Vestibular Perception and Action Employ Qualitatively Different Mechanisms: I. Frequency Response of VOR and Perceptual Responses during Translation and Tilt. Journal of Neurophysiology 94:186-198. Merfeld D, Park S, Gianna-Poulin C, Black FO, Wood S (2005b) Vestibular perception and action employ qualitatively different mechanisms: II. VOR and perceptual responses during combined Tilt & Translation. Journal of Neurophysiology 94:199-205.
Bibliography Type:	Description: (Last Updated: 04/05/2011)
Books/Book Chapters	Merfeld D. "Rotational Cues Influence Tilt and Translation Responses: Implications for Spaceflight Adaptation. " in "6th Symposium on the Role of the Vestibular Organs in Space Exploration." Ed. F. Black. in press, 2007. , Jan-2008